

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2022-23

Zoology

Paper: ZOB-401 — Fundamental Endocrinology and Economic Zoology

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A and three questions from Section-B. Question No. 1 in both sections is compulsory. Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

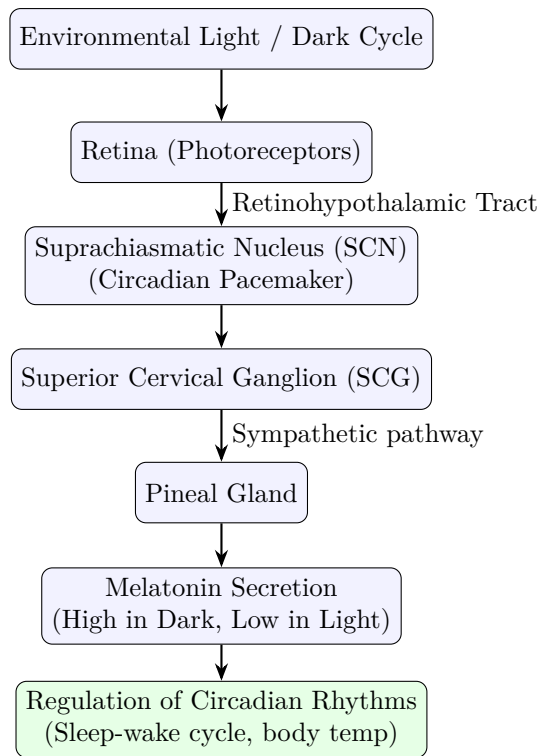
SECTION – A
(Fundamental Endocrinology) (Marks: 35)

Question 1. Answer the following parts:

- (a) Define the following terms: [1 × 3 = 3]
- (i) Bursicon
 - (ii) Cretinism
 - (iii) Relaxin
- (b) Differentiate between the following: [2 × 2 = 4]
- (i) Negative and Positive feedback mechanisms
 - (ii) Sertoli cells and Leydig cells
- (c) Justify giving reasons whether the following statements are True or False: [2 × 2 = 4]
- (i) Environmental factors affect thyroid hormone secretions.
 - (ii) Glucagon is a hypoglycemic hormone.

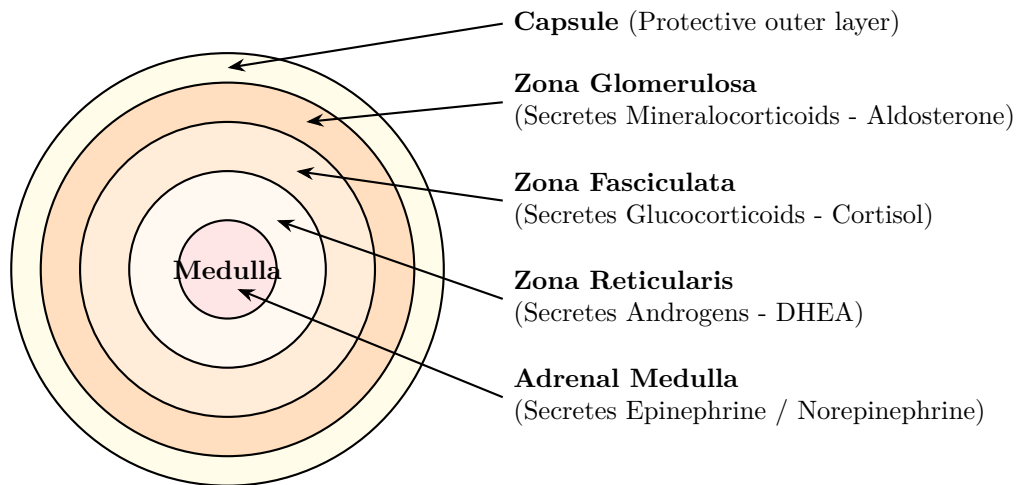
Question 2. Answer the following parts:

- (a) Define hormones and classify them according to their structural and functional similarities. [6]
- (b) How does the pineal gland coordinate daily rhythms? Explain with the help of a flowchart. [6]



Question 3. Answer the following parts:

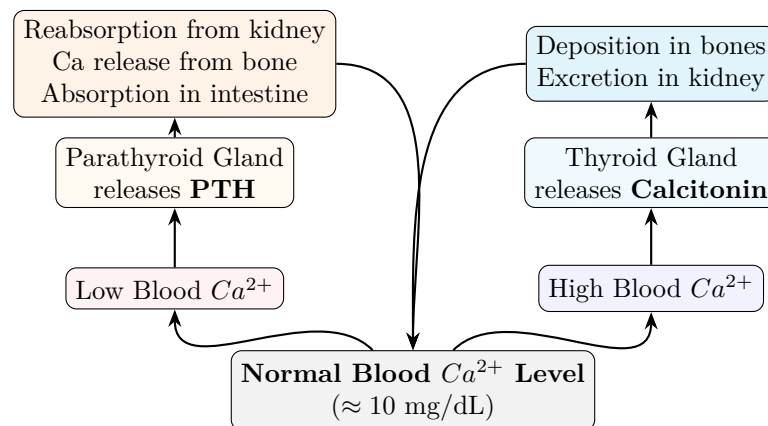
- (a) Draw a neat and well-labeled diagram of the adrenal gland. Also mention the biological actions of all the adrenocortical hormones. [8]



- (b) Describe the neurohypophyseal hormone family. [4]

Question 4. Write short notes on any two of the following: [6 × 2 = 12]

- (a) Calcium homeostasis



- (b) Graafian follicle

- (c) Ecdysteroids

SECTION – B (Economic Zoology) (Marks: 35)

Question 1. Answer the following parts:

- (a) Write whether the following statements are True or False. Justify your answers in 2-3 sentences each: [2 × 3 = 6]

- (i) Female silk moths survive about 3 years.
- (ii) Male and female lac insects reside in similar shaped resin cells.

(iii) Leghorn, Minorca and Ancona are good egg layers.

(b) Define the following terms:

[1 × 5 = 5]

(i) Pullets

(ii) Heifer

(iii) Bee-economics

(iv) Seed lac

(v) Waggle dance

Question 2. Describe the steps of fish culture, mode of breeding and methods of fish preservation. [12]

Question 3. Write an essay on milk and milk products. [12]

Question 4. Answer the following parts:

(a) Write an essay on the diseases of silkworms.

[8]

(b) Write about the economic importance of honeybees.

[4]

[Click Here](#)

[Click Here](#)

[Click Here](#)